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Study on the welding mechanism of coated glass by ultrafast laser

Tao Chen ^a, Haodong Ren ^a, Hong Shen ^{a b}   [Show more](#) ▼

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Abstract

Ultrafast laser welding is a significant area of research in joining transparent materials. Current studies primarily focus on glass-to-glass and glass-to-metal welding. However, achieving reliable connection strength while preserving good electrical performance remains a challenge in coated glass welding. This paper investigates picosecond laser welding of copper-coated glass, analyzing the effects of pulse energy and scanning speed on shear strength. Under 5μJ pulse energy and 100mm/s scanning speed, optimal welding strength (29.4MPa) is achieved. Through analysis of weld morphology and elemental distribution, three coated-glass welding mechanisms are identified. Based on the differences in the joining materials at the weld interface, these are classified as glass-glass connection, glass-copper connection, and copper-copper connection.

Outline

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Corresponding welding parameter windows were established according to each welding mechanism.

Keywords

Coated glass; Ultrafast laser; Welding mechanism; High welding strength

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1. Introduction

With the rapid development of 5G mobile communication, satellite communications, and Internet of Things (IoT) technologies, coated glass has gained widespread adoption owing to its unique advantages. Employing glass as the substrate with precisely deposited functional conductive film layers, it combines the substrate’s excellent optical transmittance, high mechanical strength, and chemical stability with the coating’s superior conductivity. This material has emerged as a critical component in fabricating next-generation antennas [1,2]. However, achieving high-performance and reliable antennas largely relies on robust and durable connections between structural components or with supporting carriers.

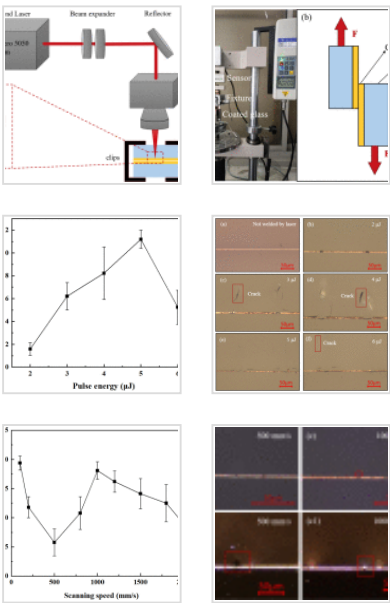
Currently, the commonly used connection techniques for glass materials mainly include adhesive bonding [3], active brazing [4], low-temperature diffusion welding [5], and laser welding [6,7]. However, each of these methods has distinct advantages and disadvantages. Active brazing usually requires the introduction of active elements at the interface to enhance the interfacial wetting performance, and it usually needs to be heated at a high temperature for a long time [8,9], which may cause the coating to fail during the brazing process. Adhesive bonding is the most commonly used connection method, which can achieve bonding between heterogeneous materials and can overcome

Data availability

References

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- Table 1. Chemical composition of K9 glass.
- Table 2. The parameters used in the wel...

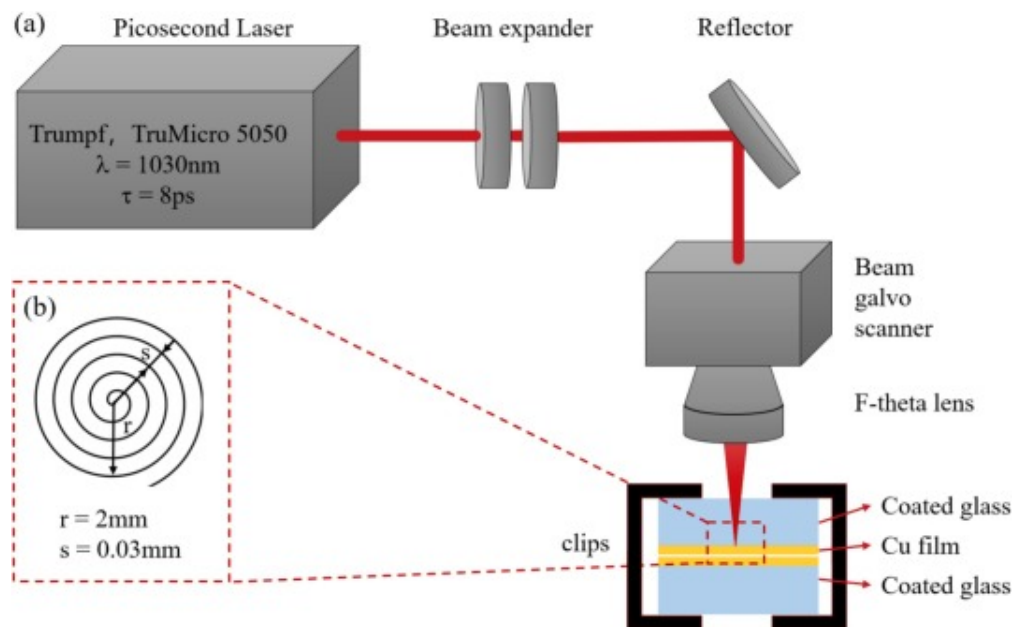
the differences in thermal expansion coefficients between materials. However, the environmental adaptability of adhesives is poor, and they are prone to aging due to external conditions such as humidity and temperature changes, affecting the long-term reliability of the bonding [10,11]. The core principle of low-temperature diffusion welding lies in achieving high-strength solid-state connections at temperatures far below the material's melting point by reducing diffusion activation energy and enhancing atomic migration capability. However, low-temperature diffusion bonding is currently primarily used for welding metals. Due to glass's extremely low diffusion capability, insulating properties, thermal mismatch, and chemical incompatibility at interfaces, low-temperature diffusion bonding is not highly suitable for bonding coated glass [12,13]. Conventional laser welding heats the metal material locally through the glass, causing the glass and metal materials to melt, generating a large molten pool, thus being suitable for welding scenarios with gaps. However, it is prone to thermal damage and cracks [14], [15], [16], and the thermal expansion coefficient of the coating material is significantly different from that of the glass, which may cause cracks or film layer detachment at high temperatures. Ultrafast laser welding, due to its short pulse time (10^{-15} – 10^{-12} s) and high peak power characteristics, can significantly reduce the heat affected zone, and the nonlinear absorption property of ultrafast laser enables it to precisely focus energy within the glass, enabling micro welding at the interface without causing thermal damage to the surrounding area [17,18], effectively inhibiting the ablation or oxidation of the coating material. Currently, in the research on laser welding of coated glass, the presence of metal film mainly serves as an intermediate absorbing layer to help the glass absorb the laser energy and thus promote the melting of the glass [19,20], which in turn enlarges the connection area at the welding interface. And most of the studies have focused on the changes in transparency of the coated glass welding area after welding [21,22], and there are few studies on the retention of the metal film function after welding.

This study investigates the effects of pulse energy and scanning speed on the weld strength of coated glass. Based on weld morphology characteristics and elemental distribution patterns, three welding mechanisms for coated-glass welding are identified

and summarized. Corresponding parameter windows are also established for different welding mechanisms and application scenarios.

2. Methodology and experimental setup

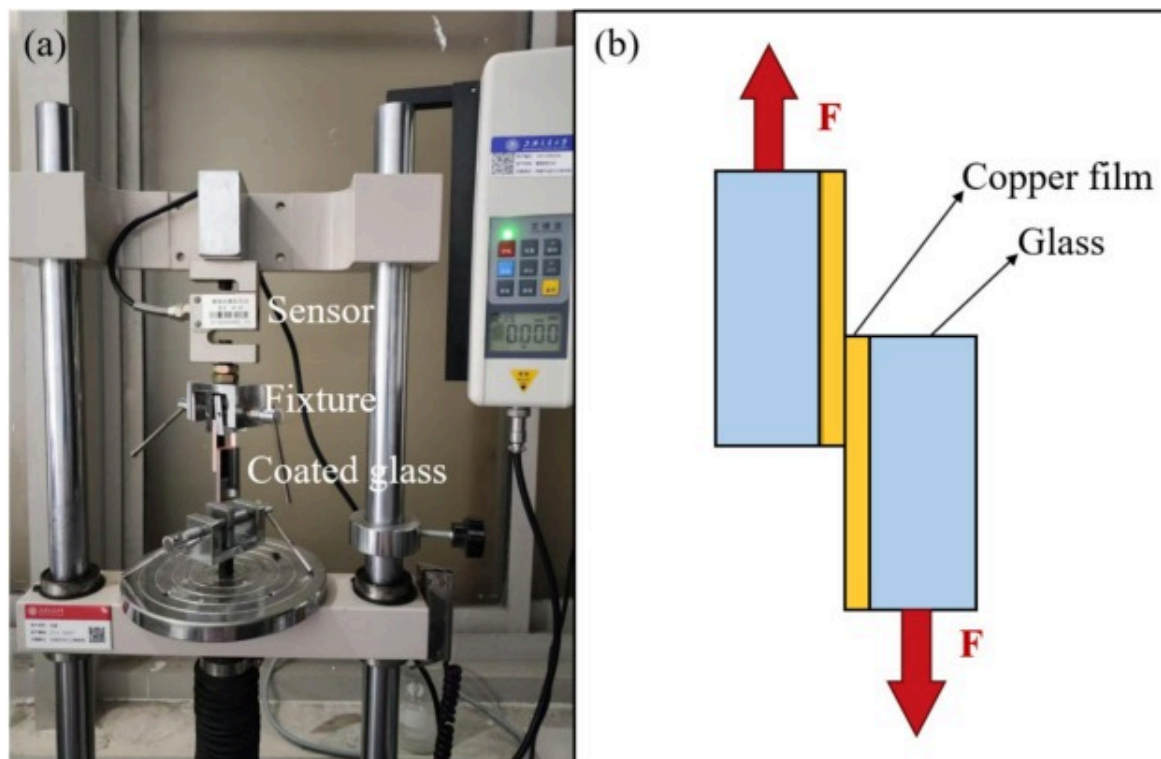
The equipment used in the experiment is the TruMicro 5050 ultra-fast laser welding system, as shown in Fig. 1(a). The laser pulse wavelength is 1030nm, and the pulse duration is 8ps. The maximum repetition frequency of the laser system is 400kHz, and the maximum pulse energy is 125μJ. The laser beam is introduced to the galvanometer through a beam expander and a reflector and then focused by an F-θ lens with a focal length of 100μm. The focused spot diameter is 20μm, and the maximum scanning speed of the laser welding system is 9000mm/s. The scanning pattern adopts a spiral line filling scan, as shown in Fig. 1(b). The scanning pattern consists of a circular contour with a radius of 2mm and internal spiral lines with an interval of 0.03mm, which is used to achieve high coverage of the welding area and avoid local overheating caused by overlapping of the welding area. To ensure consistent conditions for each welding experiment, we employ a simple mechanical fixture to secure the samples to be welded and control the clamping force between the fixtures to approximately 60N. However, due to limitations in the surface roughness of the samples, optical contact is not achieved between them. Fig. 2 shows the shear test for the welded samples.



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Fig. 1. (a) Schematic diagram of the laser system; (b) Welding path.



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Fig. 2. (a) The equipment for the tensile test; (b) Schematic diagram of the shear test.

The experiment uses K9 glass substrates with dimensions of 40mm×20mm×5 mm. The chemical composition is shown in Table 1. A pure copper film with a thickness of 2μm is deposited on the glass surface through a vacuum evaporation process. The surface roughness of the coated samples is measured using a contact-type surface roughness measuring instrument (Mitutoyo SJ-210), and the measured surface roughness of the samples is Ra=0.03μm. Prior to ultra-fast laser welding, the coated glass substrates are cleaned in an ultrasonic bath to remove particulate and organic contaminants, followed by drying under a stream of nitrogen gas. The experimental parameters used during the welding process are presented in Table 2.

Table 1. Chemical composition of K9 glass.

Composition	SiO ₂	B ₂ O ₃	BaO	Na ₂ O	K ₂ O	As ₂ O ₃
Mass fraction	69.13%	10.75%	3.07%	10.40%	6.29%	0.36%

Table 2. The parameters used in the welding process.

Parameter	Symbol	Parameter scale
Pulse energy (μJ)	E	2,3,4,5,6
Repetition frequency (kHz)	f	400
Scanning speed (mm/s)	v	100,200,500,800,1000,1200,1500,1800,2000

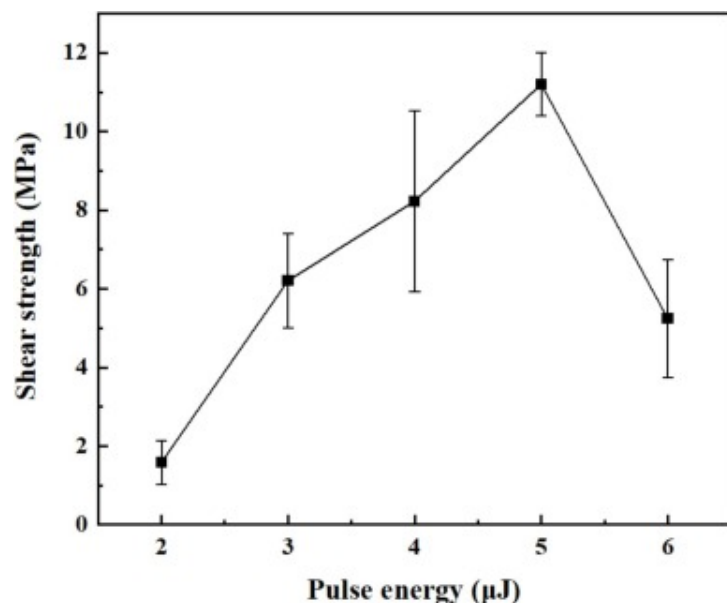
After the welding process is completed, the shear strength of the specimens is tested using an electronic universal tensile testing machine (HANDPI HDV-5K). During the test, the welding interface is kept parallel to the loading direction, and the specimen ends are fixed using fixtures to ensure that the load is uniformly applied to the welding area. The tensile rate is set at 1 mm/min. The equipment used in the experiment is the TruMicro 5050 ultra-fast laser welding system, as shown in Fig. 1(a). The laser pulse wavelength is 1030nm, and the pulse duration is 8ps. The maximum repetition frequency of the laser system is 400kHz, and the maximum pulse energy is 125μJ. The laser beam is introduced to the galvanometer through a beam expander and a reflector and then focused by an F-θ lens with a focal length of 100μm. The focused spot diameter is 20μm, and the maximum scanning speed of the laser welding system is 9000mm/s. During the experiment, the focal position of the laser pulse is set on the lower surface of the upper coated glass, and the z-axis position remained unchanged throughout the process. The scanning pattern adopts a spiral line filling scan, as shown in Fig. 1(b). The scanning pattern consists of a circular contour with a radius of 2mm and internal spiral lines with an interval of

0.03 mm, which is used to achieve high coverage of the welding area and avoid local overheating caused by overlapping of the welding area.

3. Results and discussion

3.1. Influence of pulse energy on shear strength

To systematically study the influence of pulse energy on the mechanical properties of copper-plated film glass welding joints, the experiment adopts the method of controlling variables. The scanning speed v is fixed at 1 mm/s, and the repetition frequency f is set at 400 kHz. The values of E are selected from 2 to 6 μJ (with a step size of 1 μJ). The scanning speed is fixed to control variables and eliminate its influence on shear strength, ensuring that shear strength is solely affected by pulse energy. To reduce the impact of random errors, the average value of each group of parameters is taken after three repeated experiments, and this is used as the final shear strength. The results are shown in Fig. 3. The shear strength shows a trend of increasing first and then decreasing with the increase in pulse energy. When the pulse energy is 5 μJ , the shear strength is 11.2 MPa. However, when the pulse energy further increases, the shear strength drops rapidly.



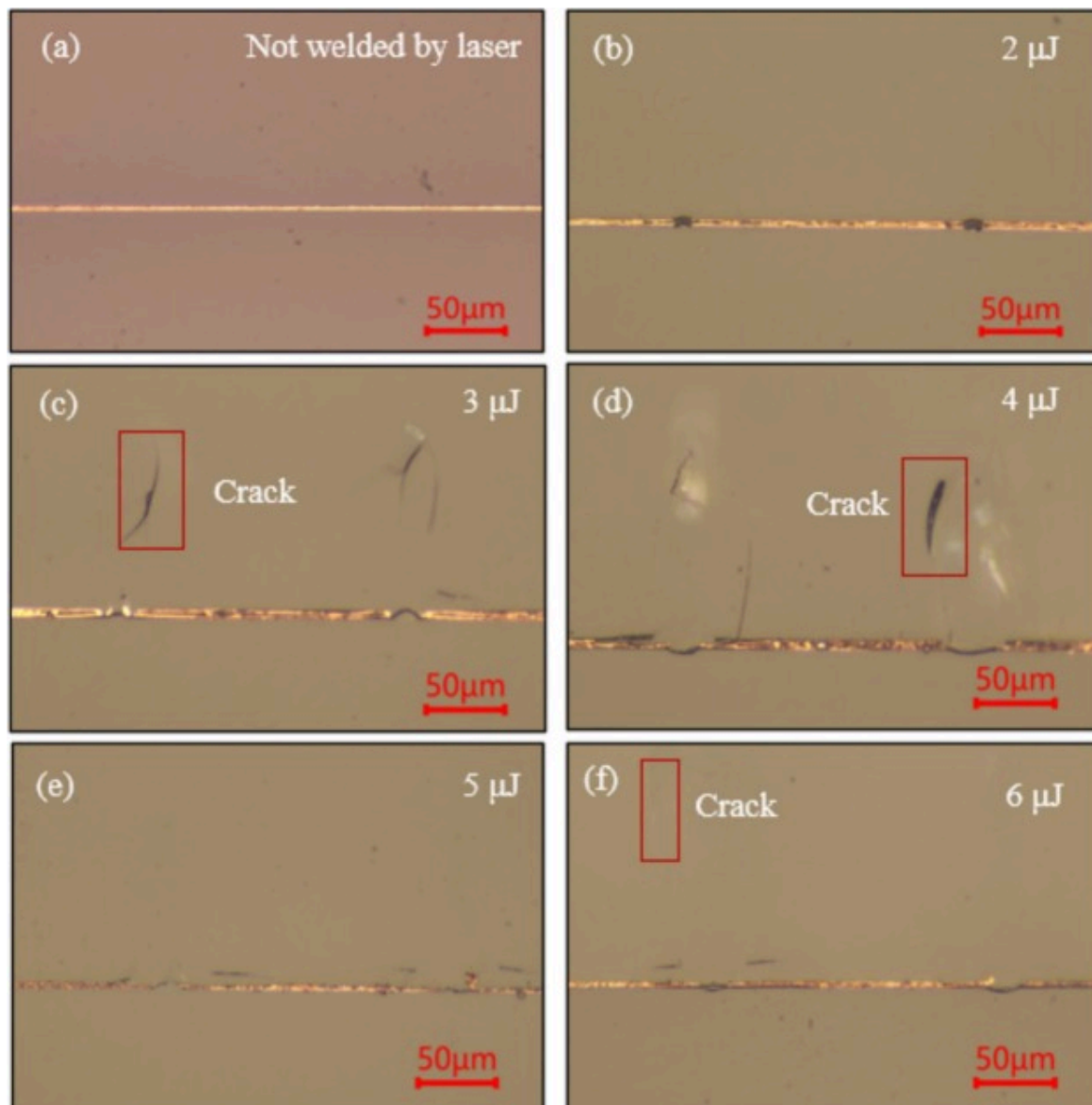
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Fig. 3. Welding strength corresponding to different pulse energies.

In this study, the cross-section of the weld of the typical sample is polished and treated. The results are shown in Fig. 4. The morphology of the unprocessed laser-coated glass is shown in Fig. 4(a). When the pulse energy is $2\mu\text{J}$, the laser energy is mainly absorbed by the copper film, causing the copper film to ablate and be removed. However, the glass interface fails to achieve effective melting due to insufficient energy input, resulting in a relatively low welding strength. When the pulse energy is 3 or $4\mu\text{J}$, the copper film vaporizes under the laser action and forms an upward-expanding plasma. The shielding effect of the plasma [23,24] significantly reduces the actual deposited energy at the welding interface. Specifically, plasma with a high free electron density absorbs most of the energy from subsequent incident laser pulses, converting it into the plasma's own thermodynamic internal energy for its maintenance and expansion rather than transferring it to the welding interface. Furthermore, plasma inhomogeneities cause

refraction and scattering of the laser, dispersing the laser energy into surrounding areas and significantly reducing the deposited energy at the interface. Consequently, most of the energy is absorbed in the glass interior, resulting in obvious defects (as indicated by the red box in Fig. 4(c)). The remaining energy penetrates the plasma, causing the glass at the interface to melt. However, due to the difference in thermal expansion coefficients between the glass and the copper material, there is a significant temperature gradient in the connection area during the laser action time, leading to a concentration of thermal stress and triggering transverse cracks, thereby affecting the welding strength.



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Fig. 4. The morphology of the weld cross-section with different pulse energies ($v=1$ mm/s) (a) Not welded by laser; (b) 2 μJ ; (c) 3 μJ ; (d) 4 μJ ; (e) 5 μJ ; (f) 6 μJ .

When the energy is increased to 5 μJ , the copper film within the laser action area is completely removed, and the glass at the interface melts to form a continuous molten area. The density of transverse cracks decreases, and the welding strength also significantly improves. However, microcracks remain present in the connection region, which limits the increase in shear strength. When the energy is increased to 6 μJ , the excessively expanded plasma reduces the energy deposited at the interface, leading to insufficient melting at the interface. Additionally, due to the plasma shielding effect, most of the laser energy is absorbed by the glass interior, generating obvious radial cracks in the glass. Owing to the magnification factor of the microscope, only the bottom end of the radial cracks is captured (as indicated by the red box in [Fig. 4\(f\)](#)).

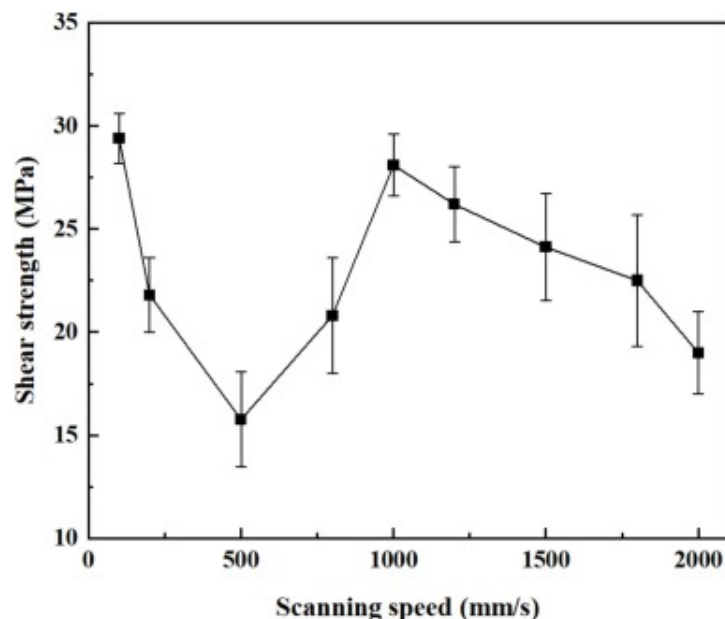
At pulse energies of 3–6 μJ , transverse microcracks of varying degrees appear across the cross-section of the coated glass. The causes of these transverse cracks are further explained as follows. During the high-energy pulsed laser interaction within the glass, a pronounced temperature gradient is generated, leading to significant thermal stresses within the glass. These stresses directly cause microcracks to form internally. Furthermore, after laser processing, high residual stresses remain within the glass. These stresses are released during subsequent grinding and polishing processes, causing cracks to propagate from micro-defects or stress concentration points within the laser-affected zone, which ultimately results in transverse microcracks.

Further elucidate the mechanisms governing the interaction between ultrafast laser and glass, as well as the role of plasma effects in influencing energy deposition at the interface. The interaction between ultrafast laser and glass is primarily dominated by nonlinear absorption mechanisms. Due to the wide bandgap characteristics of transparent materials, conventional lasers struggle to achieve energy deposition through linear absorption. However, ultrafast laser, with their extremely high peak power density, can induce nonlinear excitations within the material via multiphoton absorption and avalanche ionization, thereby forming a high-density plasma at the focal region. This plasma transfers energy to the crystal lattice through electron–phonon coupling, causing transient melting and phase transitions in the material [7]. An increase in laser energy

causes a rise in pulse energy density, promoting multi-photon ionization and avalanche ionization processes [25], leading to a growth in the initial electron density of the plasma. This creates conditions enabling the plasma to expand upward. Additionally, high energy input intensifies the energy absorption of the plasma, resulting in a rise in the temperatures of electrons and ions, and the internal pressure drives the plasma to expand outward [26]. Due to the clamping effect [27], although the pulse energy rises, the actual energy deposited at the interface does not rise. This occurs because under the influence of ultrafast lasers, the self-focusing effect and the defocusing effect induced by the plasma achieve a dynamic equilibrium, effectively imposing an upper limit constraint on the beam's peak power density. In this equilibrium state, increasing the light intensity raises the plasma density, enhancing the plasma's defocusing effect and thereby suppressing further intensity growth. Consequently, the peak power density in the beam core region is stably clamped near a nearly constant value. This process is consistent with the observed changes in the cross-sectional morphology of the weld seam in experiments.

3.2. Influence of scanning speed on shear strength

Based on the influence pattern of pulse energy on intensity and the weld morphology corresponding to different single-pulse energies, the pulse energy is fixed at 5 μ J. Based on the preliminary experiments, the selection range of scanning speed is from 100mm/s to 2000mm/s. The welding shear strength under different scanning speeds is measured, and the results are shown in Fig. 5. The shear strength shows a non-monotonic characteristic with the change of scanning speed. The shear strength obtained at scanning speeds of 100mm/s and 1000mm/s is 29.4MPa and 28.1MPa, respectively. The influence pattern of scanning speed on the welding strength exhibits a trend of first decreasing, then increasing, before slowly decreasing.

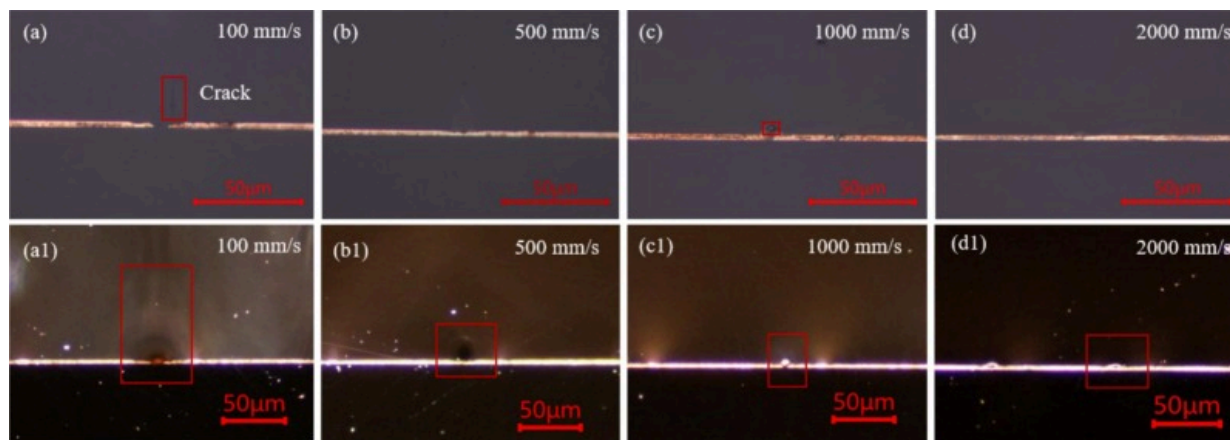


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Fig. 5. Shear strength at different scanning speeds.

The microscopic morphology of the weld cross-section of the sample under different scanning speeds is shown in Fig. 6. When the scanning speed is 100mm/s, the energy density is relatively high. Some of the laser energy is absorbed by the copper film, leading to the two layers of copper film to be completely removed. Since the copper film is completely removed, the remaining energy can be fully absorbed by the glass. Under this condition, the glass-glass direct melting connection occurs. There is no obvious temperature gradient in the connection area, so a uniform and dense connection layer is formed, as shown in Fig. 6(a). It is worth noting that excessive energy input causes the glass to absorb significant energy, generating some radial cracks inside the glass (as indicated by the red box in Fig. 6(a)). However, the mutual connection of the two layers of glass matrix materials ensures that the shear strength remains at a relatively high level.



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Fig. 6. Microstructure images of the weld captured in the bright field (upper side) and the dark field (lower side) at different scanning speeds ($E = 5 \mu\text{J}$).

Within the speed range of 500mm/s to 1000mm/s, the energy density consistently decreases as the scanning speed increases. However, high-speed scanning also inhibits the upward growth of the plasma. This occurs because as the scanning speed increases, the duration of the pulsed laser's interaction with the material decreases, resulting in reduced equivalent energy absorption per point. Plasma formation and growth require sufficient energy density and duration. During high-speed scanning, the energy density drops, inhibiting the efficiency of multiphoton ionization and avalanche ionization. Simultaneously, it disrupts the sustained energy input necessary for plasma generation and maintenance, thereby suppressing plasma growth. This inhibition allows the energy to concentrate within the focal area, making it more focused within the laser focal region rather than being absorbed or scattered by the plasma. Within this scanning speed range, the connection between the two coated glasses still depends on the bonding between the glass and the copper film. However, because the laser energy is concentrated in the focal area, the connection area at the interface increases, as shown in Fig. 6(c). This indicates that at higher scanning speeds, the concentration of energy promotes the formation of a

larger connection area at the interface, thereby enhancing the welding shear strength. Notably, although the energy density decreases, some defects caused by ablation appear inside the glass due to energy concentration (as indicated by the red box in [Fig. 6\(c\)](#)).

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Within the speed range of 1000mm/s to 2000mm/s, the energy density continuously decreases. The copper in the laser-affected area no longer undergoes ablation but instead melts under the action of laser energy. The connection method also changes to a melting connection between two layers of copper films, as shown in [Fig. 6\(d\)](#). As the scanning speed increases, the heat input decreases continuously, the thermal accumulation effect diminishes, and the molten pool progressively shrinks. Correspondingly, the mechanical properties of the welded joint exhibit a negatively correlated attenuation with the scanning speed.

To evaluate the welding area in relation to the area softened by laser radiation, the cross-sectional morphology of the sample under dark field conditions is collected, as shown in [Fig. 6\(a1-d1\)](#). As can be seen from the figure, as the scanning speed increases, the laser thermal affected zone gradually decreases. When the scanning speed is 100mm/s, the area softened by laser radiation is the largest. As the scanning speed increases, the area

softened by laser radiation gradually decreases. At 2000mm/s, almost all the laser energy acts on the interface of the connection. Therefore, the welding area is almost the same as the laser radiation softened area. It is worth noting that under the dark-field imaging condition, it can be observed that on the side of each laser action area, there is an additional laser action trace. This is because the laser was not turned off in time during the laser scanning process.

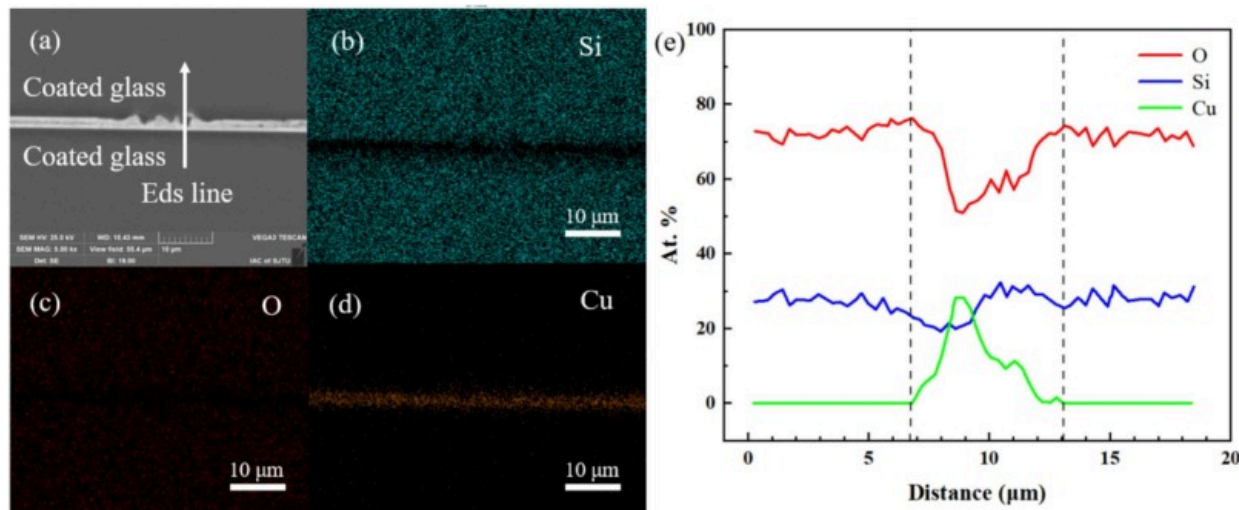
In summary, both low and high speed scanning result in good welding results, while the strength remains poorer at scanning speeds around 500mm/s. Low-speed scanning is conducive to the melting of the material, and it ensures a uniform temperature distribution in the welded area, thus producing a high-quality welded joint. When scanning at high speed, the energy concentrates at the weld interface, which is conducive to increasing the connection area at the interface and thus obtaining good weld strength.

3.3. Welding mechanism of coated glass

Based on the differences in the bonding materials at the interface after welding, the welding mechanisms can be classified into three typical types. The mechanisms and interface characteristics for each type are analyzed as follows.

In mechanism 1, the copper film is partially ablated by the laser, generating voids. The glass subsequently melts under subsequent laser action and flows into the voids formed due to the ablation of the copper film, ultimately achieving a fusion connection between the glass and the copper, as shown in Fig. 7. The SEM image reveals that the interface exhibits a typical interlocking structure. The molten glass flows downward toward the copper film side, while the melted copper infiltrates the interior of the glass, forming a composite layer with a depth of approximately 6 μ m. This interlocking structure with mechanical interlocking features significantly strengthens the mechanical anchoring strength of the interface. The EDS line scan results indicate that the Si element in the interface area maintains a continuous distribution. This phenomenon likely occurs

because the line scan path region partially achieves a connection between the glass, leading to the Si element content not decreasing to zero at that location.

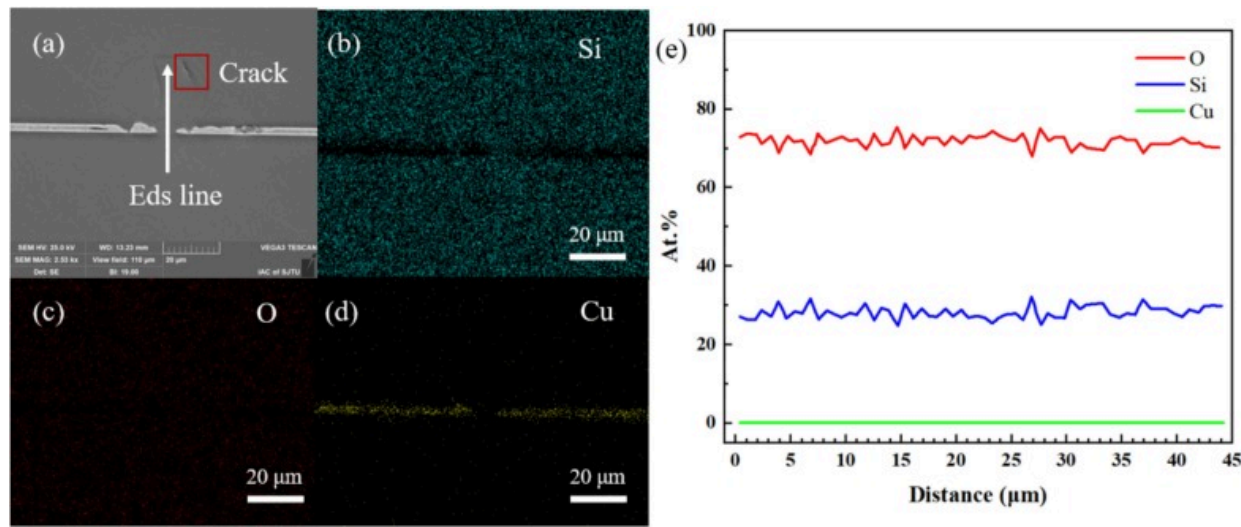


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Fig. 7. (a) SEM image of the welded cross-section corresponding to Mechanism 1; (b-d) Element distribution of the welded cross-section; (e) Elemental line distribution.

In mechanism 2, the copper film is completely ablated under the action of the laser energy, and the formed plasma mixes completely with the glass plasma, forming a weld after cooling, which presents a glass-glass connection due to the low copper content. The SEM image shows that the upper and lower glass substrates achieve a direct fused connection, and the Si element interface exhibits a continuous distribution, while the Cu element content remains at 0, as shown in Fig. 8. This confirms that the gap between the materials is entirely filled with the molten glass. It is worth noting that the lack of material inside the glass due to the molten glass filling the voids created by the ablation of the copper film causes microcracks to appear inside the glass(as indicated by the red box in Fig. 8(a)).

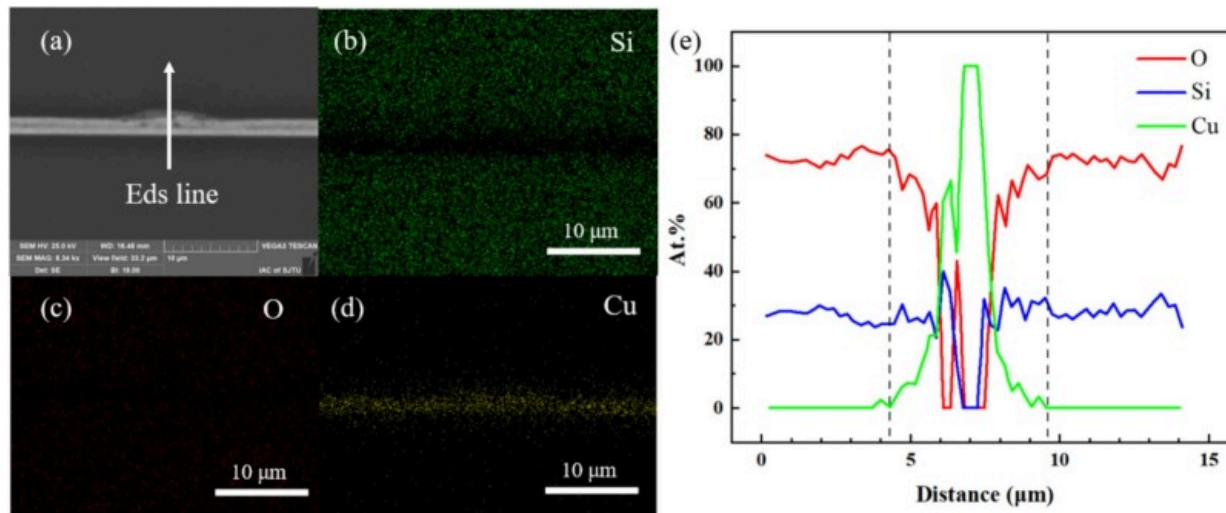


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Fig. 8. (a) SEM image of the welded cross-section corresponding to Mechanism 2; (b-d) Element distribution of the welded cross-section; (e) Elemental line distribution.

In mechanism 3, due to the lower energy deposited at the interface, the copper film does not undergo ablation removal at this time but instead melts under the laser effect. The first layer of copper film melts first, while the second layer of copper film gradually melts under the influence of heat conduction. Under the subsequent laser pulse action, the molten pool gradually expands, ultimately achieving the melting and bonding of the two layers of copper film. The SEM image shows that the melted copper penetrates the adjacent glass substrate, with a penetration depth of approximately $1.2 \pm 0.3 \mu\text{m}$, as shown in Fig. 9. The element distribution indicates that a Cu element-rich area with a width of about $5 \mu\text{m}$ is formed in the connection area, indicating that this interface is completely formed by the melting of the copper film. This mechanism is significantly different from the previous two types. This mechanism achieves connection through the melting of two layers of copper films.

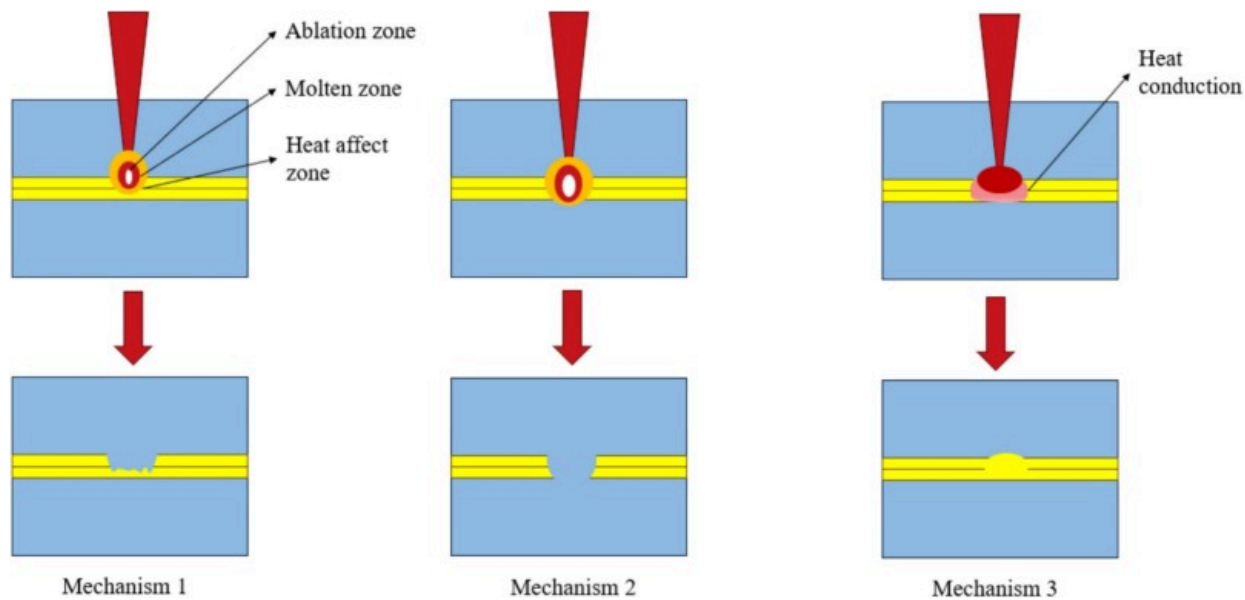


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Fig. 9. (a) SEM image of the welded cross-section corresponding to Mechanism 3; (b-d) Element distribution of the welded cross-section; (e) Elemental line distribution.

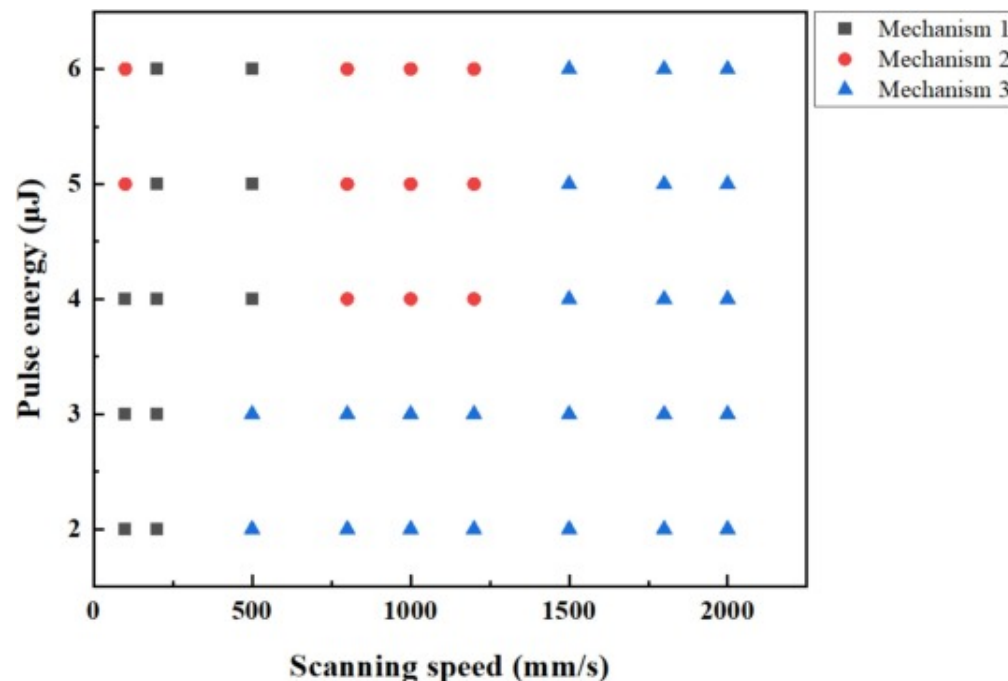
The corresponding welding schematic diagrams for the three mechanisms are shown in Fig. 10. The experiment further expands the range of welding parameters, conducts weld interface analysis under the corresponding parameters, and classifies the welding mechanisms based on the morphological characteristics. As a result, the parameter windows for different mechanisms are obtained, as shown in Fig. 11.



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Fig. 10. Schematic diagram of the coated glass welding mechanism.



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Fig. 11. (a) Parameter windows corresponding to the three welding mechanisms.

Mechanism 1 achieves a heterogeneous connection between glass and copper while retaining some copper layers. It is suitable for applications requiring local conductivity, and its advantage lies in being able to balance the interface connection strength and the functionality of specific areas. This type has moderate welding efficiency, but it requires precise control of the copper film removal accuracy and the thermal influence range. Mechanism 2 completely removes the copper film and causes the glass substrate to melt to achieve a connection between the glasses. Its characteristic is that it achieves a high-strength glass-to-glass connection, but usually requires a lower scanning speed, that is, a longer laser-material interaction time to ensure that the glass fully melts and fills the gaps. It is suitable for applications requiring high-strength connections. Mechanism 3 involves the melting of copper films to achieve mutual connection. Mechanism 3 achieves

a bonding method where the copper coating between two glass plates fuses together, meaning the copper coating on the coated glass is preserved without being ablated or removed. While maintaining the complete copper film structure, it also has the highest welding efficiency and is suitable for scenarios where the copper film functionality needs to be retained.

4. Conclusions

This study systematically explores the influence of pulse energy and scanning speed on the interface bonding performance during the ultrafast laser welding of copper-coated glass. Through optimized parameters, a maximum welding shear strength of 29.4MPa is achieved. Three mechanisms of the welding of copper-coated glass are identified, and these mechanisms are verified by observing and analyzing the weld morphology and the element distribution at the interface. Finally, parameter windows for different welding mechanisms are summarized, providing a theoretical basis for process optimization of ultrafast laser welding of copper-coated glass in different application scenarios.

CRediT authorship contribution statement

Tao Chen: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Haodong Ren:** Writing – review & editing, Visualization, Methodology, Investigation, Conceptualization. **Hong Shen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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